

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards (BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5	Total
Marks:25	

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and indepth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples

Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

COMPARATIVE ANALYSIS OF NETWORK TOPOLOGIES

INTRODUCTION

Network topology refers to the physical or logical arrangement of computers, network devices, cables, and communication links in a network. Different topologies provide different levels of cost, reliability, performance, scalability, complexity, and maintenance.

The major topologies considered in this comparison are Star, Bus, Mesh, and Hybrid topology. Each topology is suitable for different situations depending on the size of the network, budget, performance requirements, reliability needs, and future expansion.

The following comparative analysis evaluates these topologies based on cost, reliability, performance, fault tolerance, installation complexity, scalability, and maintenance.

1. STAR TOPOLOGY VS BUS TOPOLOGY – COST, RELIABILITY AND PERFORMANCE

1.1 Cost Comparison

Star topology is generally more expensive than bus topology because every computer or network device is connected individually to a central switch or hub. This requires more network cables and a central networking device.

In a star topology, each node has a separate connection to the central switch. Therefore, the amount of cabling required increases as more devices are added. A managed switch may also increase the initial cost, but it provides better network management and monitoring.

In contrast, bus topology uses a single main backbone cable to connect multiple devices. Devices share the same communication medium, so fewer cables and networking devices are required. This makes bus topology cheaper for very small and simple networks.

Example

For a small classroom with a few computers:

Bus topology: Requires one backbone cable, connectors, and terminators.

Star topology: Requires a switch and separate cables from every computer to the switch.

Therefore, bus topology can have a lower initial installation cost.

However, modern networks generally prefer star topology because the additional cost provides better performance, reliability, management, and future expansion.

1.2 Reliability Comparison

Star topology provides better reliability than traditional bus topology.

In star topology, every device has its own individual connection to the central switch. If the cable connecting one computer fails, only that particular computer is disconnected. Other computers can continue communicating normally.

For example, if Computer 3 has a damaged network cable, Computers 1, 2, 4, and 5 can continue using the network.

In bus topology, all devices depend on a common backbone cable. If the backbone cable is broken or improperly terminated, communication across the network can be seriously affected or completely interrupted.

Therefore:

Star topology → Failure of one node/cable usually affects only that node.

Bus topology → Backbone failure can affect the entire network.

1.3 Performance Comparison

Star topology normally provides better network performance, especially when a network switch is used.

In a switched star network, the switch intelligently forwards Ethernet frames toward the required destination. Dedicated links between devices and the switch also reduce the problems associated with a shared communication medium.

Bus topology uses a shared backbone. When several devices attempt to communicate at the same time, they compete for the same medium. This can result in collisions in traditional shared Ethernet implementations.

As the number of devices increases, traffic on the shared bus also increases. Consequently, network performance can

decrease.

Performance Comparison

Factor	Star Topology	Bus Topology
Initial cost	Higher	Lower
Cabling	More	Less
Central device	Required	Not required
Reliability	High	Lower
Performanc e	Better	Decreases with traffic
Fault isolation	Easy	Difficult
Expansion	Easier	Limited
Modern usage	Very common	Mostly legacy

1.4 Overall Analysis

Although bus topology is cheaper, star topology provides a better balance between performance, reliability, management, and scalability.

Therefore, bus topology may be suitable for a very small or legacy network where cost is the main concern, while star topology is more appropriate for modern offices, schools, laboratories, and businesses.

2. MESH TOPOLOGY VS STAR TOPOLOGY – FAULT TOLERANCE

2.1 Fault Tolerance in Mesh Topology

Mesh topology provides a very high level of fault tolerance because devices have multiple communication paths.

In a full mesh, each node is directly connected to every other node. In a partial mesh, important devices have multiple connections while less-critical devices may have fewer connections.

If one communication link fails, data can potentially use another available path.

Example

Suppose a network contains four important routers:

Router A $\leftrightarrow$ Router B

Router A $\leftrightarrow$ Router C

Router B $\leftrightarrow$ Router D

Router C $\leftrightarrow$ Router D

If the connection between Router A and Router B fails, traffic can potentially be redirected through Router C and Router D. This makes mesh topology suitable for environments where continuous communication is extremely important.

2.2 Fault Tolerance in Star Topology

Star topology provides moderate fault tolerance.

The main advantage is that each node has an individual connection to the central switch. If one workstation or one connecting cable fails, the remaining devices are normally unaffected.

However, the central switch is a critical point.

If the central switch fails:

Computer A $\rightarrow$ X $\rightarrow$ Computer B

Communication between devices connected through that switch may stop.

Therefore, the central device represents a potential single point of failure unless redundancy is introduced.

2.3 Comparison

Feature	Mesh Topology	Star Topology
Number of paths	Multiple	Usually one path through central switch
Fault tolerance	Very high	Moderate
Single point of failure	Low in redundant designs	Central switch can be one
Reliability	Very high	High
Cost	Very high	Moderate
Complexity	High	Low to moderate
Maintenance	Difficult	Easier
Suitable for	Critical networks	Offices, schools, labs

2.4 Overall Analysis

Mesh topology is more robust than star topology because it provides multiple communication paths. It is particularly useful for banking systems, data centers, backbone networks, telecommunications, and other critical environments.

Star topology is simpler and cheaper and is usually preferred for ordinary LANs.

Thus, the choice depends on the importance of continuous connectivity:

Critical communication → Mesh or redundant/partial mesh

Normal LAN → Star topology

3. BUS TOPOLOGY VS MESH TOPOLOGY – INSTALLATION COMPLEXITY

3.1 Installation of Bus Topology

Bus topology is relatively simple to install because it uses a common backbone cable.

Traditional bus Ethernet networks used coaxial cable with appropriate connectors and terminators. Each computer was connected to the backbone.

A simple representation is:

Terminator → PC1 → PC2 → PC3 → PC4 → Terminator

Only a limited amount of cable and hardware is required.

Therefore, bus topology has low installation complexity for small networks.

3.2 Installation of Mesh Topology

Mesh topology is considerably more complex.

In a full mesh, every node needs a direct connection to every other node. As the number of nodes increases, the number of required links increases rapidly.

For n devices, a full mesh requires:

Number of links = $n(n - 1) / 2$

For example:

4 devices → 6 links

5 devices → 10 links

10 devices → 45 links

20 devices → 190 links

This demonstrates why full mesh becomes difficult and expensive as the network grows.

3.3 Installation Comparison

Factor	Bus Topology	Mesh Topology
Cabling	Low	Very high
Installation	Simple	Complex
Number of connections	Few	Many
Equipment requirement	Low	High
Initial cost	Low	High
Troubleshooting	Can be difficult	Complex
Scalability	Poor	Good technically, but expensive
Suitable for	Small/legacy networks	Critical networks

3.4 Practical Analysis

Bus topology is simple and economical, but it has significant limitations. A backbone cable failure can affect multiple devices, and troubleshooting can be difficult.

Mesh topology provides excellent redundancy but requires much more equipment, cabling, configuration, and maintenance.

Therefore, bus topology is easier to install, while mesh

topology provides better reliability at the cost of greater complexity.

It is important to note that traditional coaxial bus Ethernet is now largely obsolete. Modern LANs commonly use switched star topology instead.

4. HYBRID TOPOLOGY VS STAR TOPOLOGY – SCALABILITY

4.1 Scalability of Hybrid Topology

Hybrid topology combines two or more network topologies to create a network that matches the requirements of an organization.

For example, a university may use:

- Fiber-based redundant links for the backbone

- Star topology inside individual buildings

- Wireless networks for mobile users

- Additional redundant links for critical areas

Because different sections can use different topologies, the network can be expanded according to local requirements.

Example

A university initially has three buildings:

Building A + Building B + Building C

Later, Building D and Building E can be added to the existing backbone without completely redesigning the network.

This makes hybrid topology highly flexible.

4.2 Scalability of Star Topology

Star topology is also scalable, but expansion depends on the capacity of the central networking device.

For example, a 24-port switch can directly support a limited

number of connected devices. When the available ports are exhausted, another switch can be added.

The network can then be expanded into a larger hierarchical or extended-star structure.

However, adding multiple switches introduces additional equipment and management requirements.

4.3 Scalability Comparison

Feature	Hybrid Topology	Star Topology
Flexibility	Very high	High
Expansion	Highly flexible	Requires additional switches
Network design	Complex	Simple
Large organizations	Very suitable	Suitable with hierarchy
Cost	Can be high	Moderate
Management	More complex	Easier
Adaptability	Excellent	Good

4.4 Overall Analysis

Hybrid topology provides greater flexibility because different parts of an organization can use different network designs.

For example, a university, hospital, or large company can use different topologies according to building layout, security requirements, traffic requirements, and reliability needs.

Star topology is easier to expand for small and medium networks, but large organizations may use multiple

interconnected star networks as part of a hierarchical or hybrid design.

Therefore:

Small/medium network → Star topology

Large and complex organization → Hybrid topology

5. MAINTENANCE COMPARISON – STAR, MESH, BUS AND HYBRID

5.1 Maintenance of Star Topology

Star topology is relatively easy to maintain because each device has an individual connection to the central switch.

If one computer loses network connectivity, the administrator can check:

1. The computer's network adapter
2. The Ethernet cable
3. The switch port
4. IP configuration
5. Switch configuration

Since the connections are independent, faults can usually be isolated quickly.

For this reason, star topology is widely used in offices, schools, computer laboratories, and small businesses.

5.2 Maintenance of Mesh Topology

Mesh topology is more difficult to maintain because there are many links and devices.

An administrator may need to monitor:

Multiple physical links

Routing paths

- Router interfaces
- Routing protocols
- Redundant connections
- Link failures
- Traffic conditions

Although the network can continue operating after a link failure, identifying and repairing the failed link may require skilled network administrators.

Therefore, mesh provides high reliability but increases maintenance complexity.

5.3 Maintenance of Bus Topology

Bus topology can be difficult to troubleshoot because all devices share the same backbone.

A fault in the backbone cable, connector, or termination can affect multiple devices.

The administrator may need to check several points along the cable to identify the source of the problem.

This makes troubleshooting more time-consuming compared with a switched star network.

5.4 Maintenance of Hybrid Topology

Hybrid topology has moderate to high maintenance complexity because it combines different topologies and technologies.

For example, an organization may have:

- Star networks in offices
- Fiber links between buildings
- Redundant links in the backbone

Wireless access points

VLANs

Routers and firewalls

Each component may require different configuration and troubleshooting procedures.

However, hybrid topology provides flexibility and can be designed so that failures are isolated to specific sections.

5.5 Maintenance Comparison Table

Factor	Star	Mesh	Bus	Hybrid
Maintenance difficulty	Low	High	Moderate/ High	Moderate/ High
Fault identification	Easy	Complex	Difficult	Moderate
Troubleshooting	Simple	Complex	Difficult	Depends on design
Skilled administrator required	Low/ Moderate	High	Low/ Moderate	Moderate/ High
Reliability	High	Very high	Low/ Moderate	High
Flexibility	High	High	Low	Very high

5.6 Overall Analysis

Star topology is generally the easiest to maintain because each connection can be independently monitored and repaired.

Mesh topology provides excellent reliability but requires more technical knowledge because of its multiple links and routing paths.

Bus topology is simple in design but can be difficult to troubleshoot when the shared backbone develops a fault.

Hybrid topology provides the greatest flexibility, but its maintenance depends on the combination of technologies used.

OVERALL COMPARATIVE ANALYSIS

The four network topologies have different strengths and weaknesses.

Criteria	Star	Bus	Mesh	Hybrid
Cost	Medium	Low	Very High	Medium/High
Reliability	High	Low/Medium	Very High	High
Performance	High	Low/Medium	High	High
Fault tolerance	Medium	Low	Very High	High
Installation	Easy	Easy	Difficult	Moderate/Difficult
Scalability	Good	Poor	Limited by cost	Excellent
Maintenance	Easy	Difficult	Difficult	Moderate/

ce Good Low t Difficult

CONCLUSION

The comparison shows that no single network topology is best for every situation. The appropriate topology depends on the organization's budget, network size, performance requirements, reliability requirements, scalability, and maintenance capability.

Star topology provides a good balance of cost, performance, reliability, scalability, and easy maintenance, making it the most common choice for modern LANs.

Bus topology has a low installation cost and simple structure, but its shared medium, fault-isolation problems, and limited scalability make it unsuitable for most modern networks.

Mesh topology provides the highest fault tolerance and reliability because multiple communication paths are available. However, it requires more cables, ports, devices, configuration, and maintenance, resulting in higher cost and complexity.

Hybrid topology provides the greatest flexibility and scalability because different topologies can be combined according to the requirements of different parts of an organization. It is especially useful for large organizations such as universities, banks, hospitals, and corporate networks.

Therefore, the final selection should be based on the specific application:

Low-cost small/legacy network → Bus

Normal office or laboratory LAN → Star

Critical/high-availability network → Mesh or redundant partial mesh

Large and complex organization → Hybrid

Overall, modern network design commonly favors switched star and hierarchical/hybrid architectures, while mesh principles are often introduced where redundancy and high availability are essential.